

# Using GitHub For CAD

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Latest Revision: 01/25/2026

# What is GitHub?



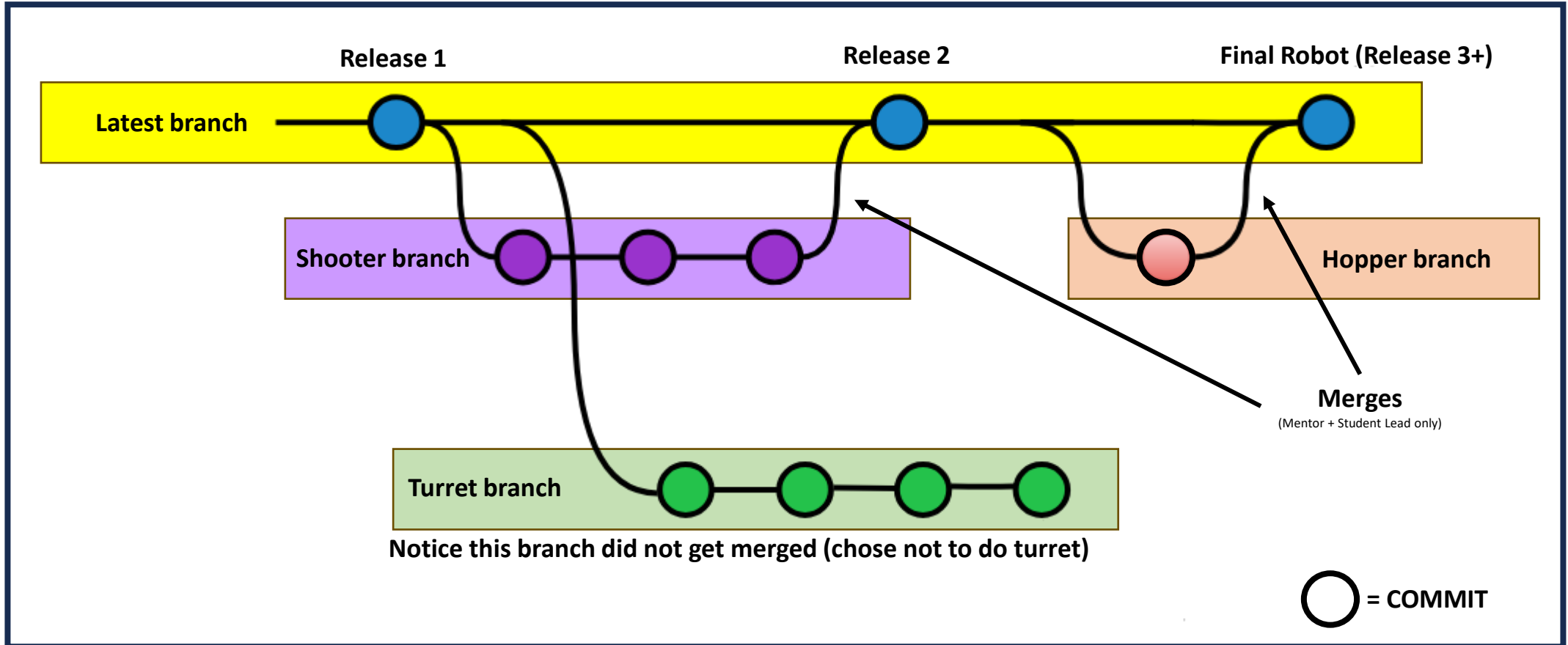
**GitHub** is an online platform used to store, share, and collaborate on software and other projects using the “Git” version control system.

GitHub brings with it some new terms you may not have heard before:

- **Repository (repo):** A collection of the entire project and its full history
- **Branch:** A parallel version of the project files, allows you to develop/test changes without affecting the main project files
- **Commit:** A saved snapshot of changes with a message explaining what was done
- **Push:** Used to push your local changes to your branch
- **Pull / Fetch:** Used to pull the latest changes from the branch you’re working on
- **Pull Request:** Notifies the admins/mentors/student sub-team lead that your branch is ready to be implemented into the main branch.  
Your changes will be added assuming it has been approved.
- **Merge:** Combining changes from one branch into another
- **Issue:** A tool in GitHub that lets you track tasks and problems within your project

# How GitHub Works

## Repository



# How Mechanical Will Use GitHub

- We are going to follow a standard process on managing files and changes
- Reference this guide whenever you are unsure

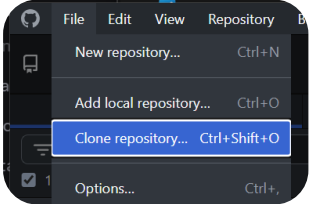
The current process:



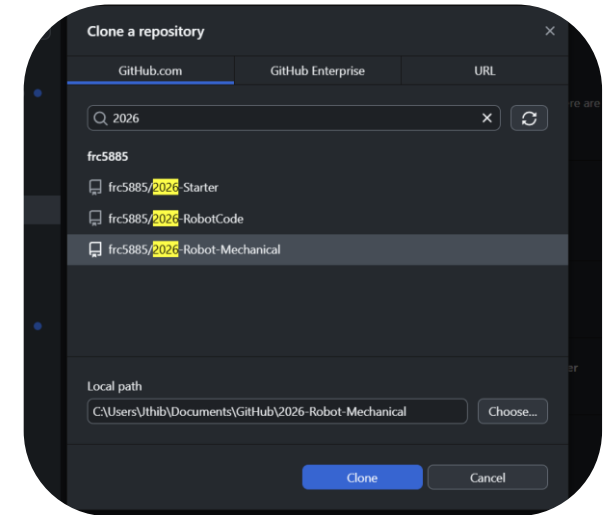
- The next few slides will guide you through using GitHub Desktop and show you which part of the process you are in

# Starting with GitHub Desktop

- Open GitHub Desktop and go to File > Clone Repository

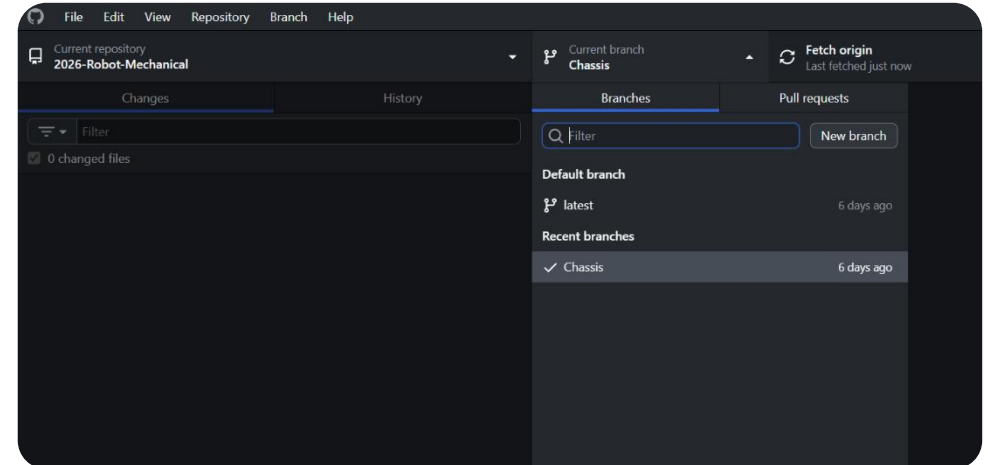


- Select the repository we are using for your project (2026-Robot-Mechanical) and select a file path (try to keep the file path consistent)
- Click “Clone” to sync the repository with the folder on your computer



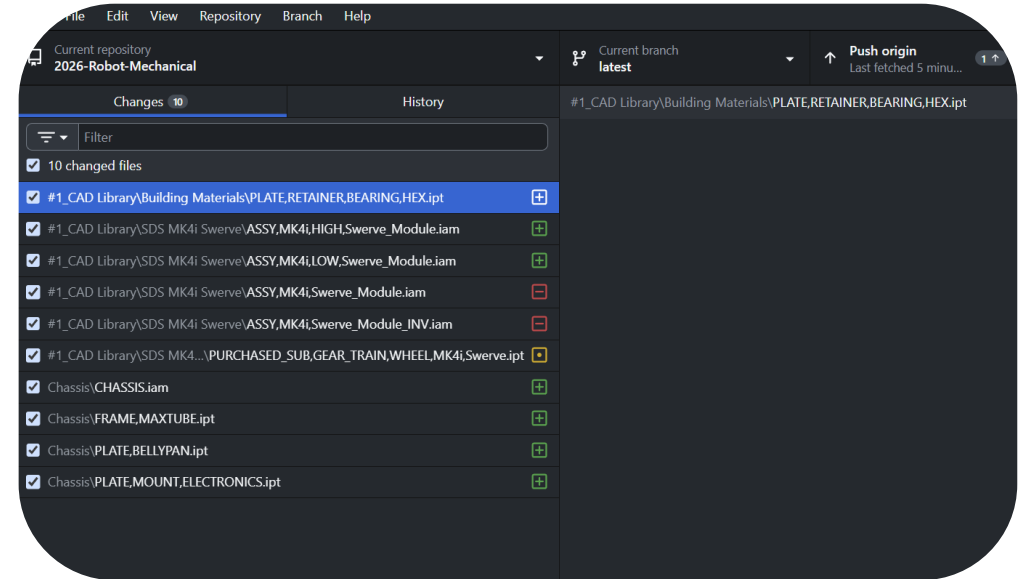
# Starting with GitHub Desktop

- Once the repo has been cloned, make sure you “Fetch Origin” to get the latest updates
- If you’re current branch is shown as latest/main/default, then select or create your branch to start work on it
- If you’ve created a new branch, make sure to publish it (it will prompt you to)
- If someone else is working on a branch, you won’t be able to edit it (it will be checked-out)



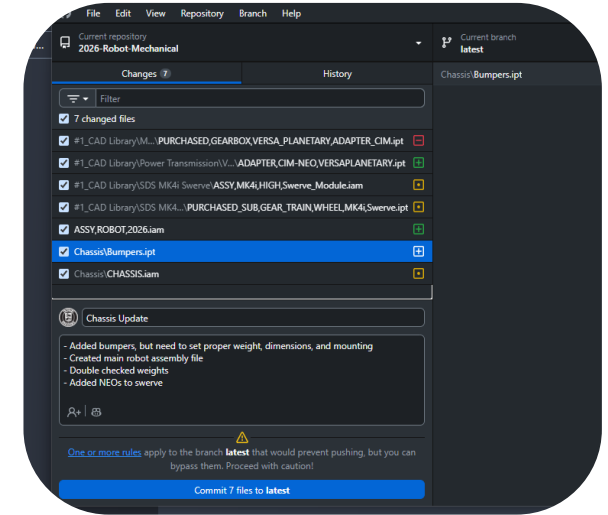
# Starting with GitHub Desktop

- Create/modify/remove files in Inventor
- Make sure you save your changes often!
- When your done your work for this session, make a commit of your changes



# Starting with GitHub Desktop

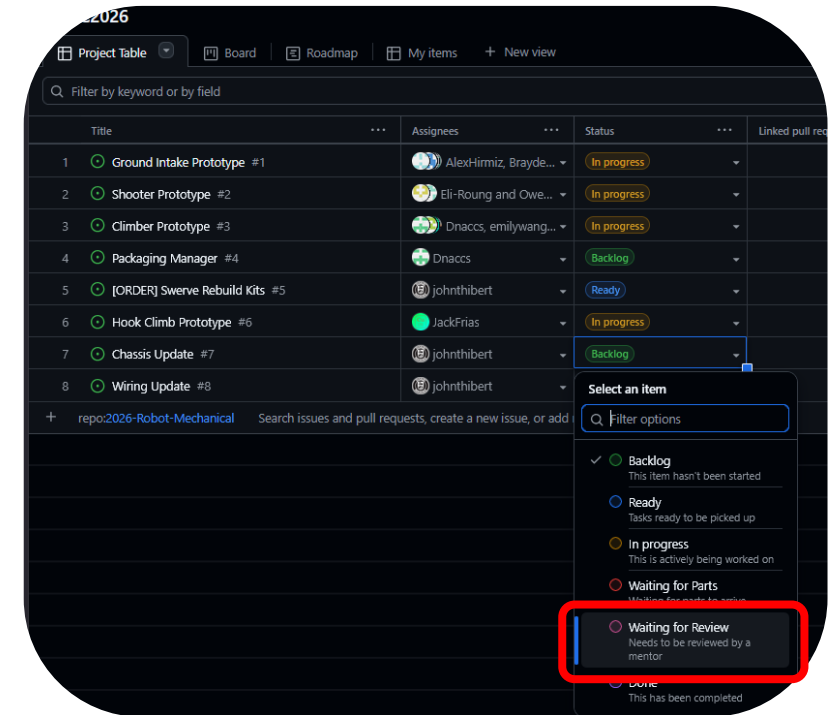
- Write a generic title for the commit
- Write a detailed but concise commit description (don't just use AI)
- Ensure you're committing to the correct branch and make sure your CAD files are saved before committing!!
- After commit, the changes are only local until you push them. Push your commits often, and make sure when your done your part, to push the changes so others can work on them



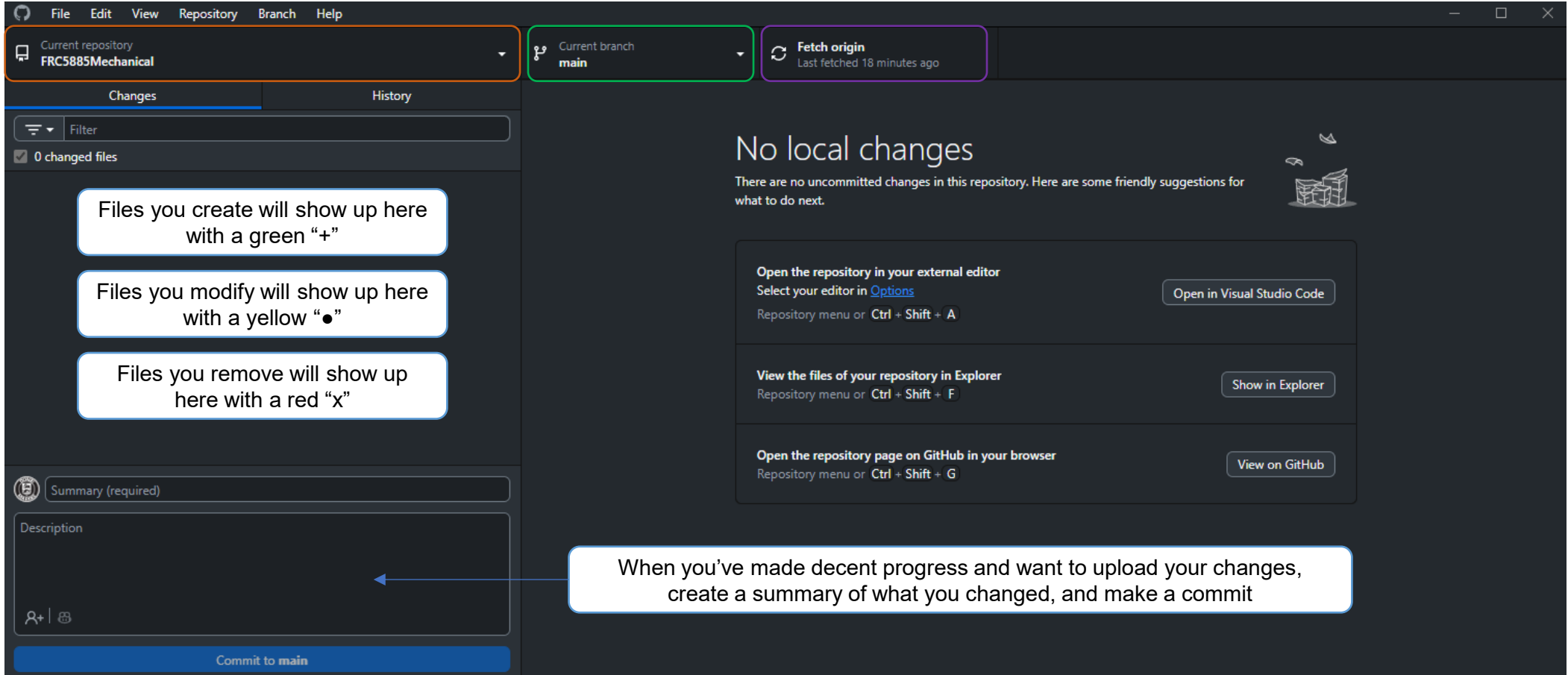


# Starting with GitHub Desktop

- Once you (and others) are done working on a branch:
  - Set the “status” tag of the task/issue to “waiting for review”
  - Contact the mechanical student leader
  - If the student lead is not available, contact a mechanical mentor
- The student lead/mentor will then review the design and may contact you to make some changes to your design (after which you will send it for review again)
- When the student lead/mentor is approving of the design, they will manually merge the changes you made in your branch with the main/latest branch



# GitHub Desktop



The screenshot shows the GitHub Desktop application window. The top bar includes a menu (File, Edit, View, Repository, Branch, Help) and three status boxes: 'Current repository: FRC5885Mechanical' (highlighted with an orange box), 'Current branch: main' (highlighted with a green box), and 'Fetch origin: Last fetched 18 minutes ago' (highlighted with a purple box). The left sidebar has a 'Changes' tab selected, showing '0 changed files'. Three callout boxes are present: 'Files you create will show up here with a green "+"', 'Files you modify will show up here with a yellow "●"', and 'Files you remove will show up here with a red "X"'. At the bottom of the sidebar is a 'Commit to main' button. The main area displays 'No local changes' with suggestions for next steps. A callout box points to the 'Summary (required)' field in the commit section, stating: 'When you've made decent progress and want to upload your changes, create a summary of what you changed, and make a commit'.

Current repository: FRC5885Mechanical

Current branch: main

Fetch origin: Last fetched 18 minutes ago

Changes

History

Filter

0 changed files

Files you create will show up here with a green "+"

Files you modify will show up here with a yellow "●"

Files you remove will show up here with a red "X"

Summary (required)

Description

Commit to main

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

Open the repository in your external editor  
Select your editor in [Options](#)  
Repository menu or **Ctrl + Shift + A**

Open in Visual Studio Code

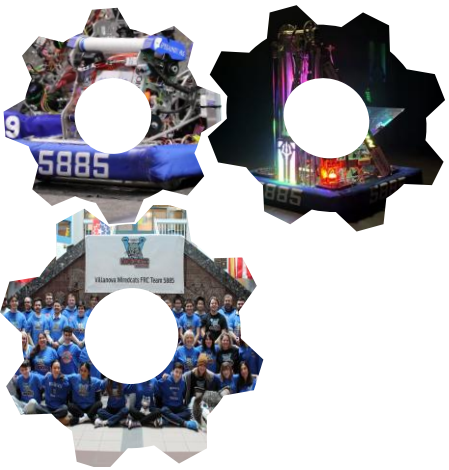
View the files of your repository in Explorer  
Repository menu or **Ctrl + Shift + E**

Show in Explorer

Open the repository page on GitHub in your browser  
Repository menu or **Ctrl + Shift + G**

View on GitHub

When you've made decent progress and want to upload your changes, create a summary of what you changed, and make a commit



# Open Discussion



[wiredcats5885.ca](http://wiredcats5885.ca)

| Revision Table    |                      |
|-------------------|----------------------|
| Date (MM/DD/YYYY) | Description          |
| 01/25/2026        | Presentation release |
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